

Elementary Students' Views of Explanation, Argumentation, and Evidence, and Their Abilities to Construct Arguments Over the School Year

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Abstract: Science includes more than just concepts and facts, but also encompasses scientific ways of thinking and reasoning. Students' cultural and linguistic backgrounds influence the knowledge they bring to the classroom, which impacts their degree of comfort with scientific practices. Consequently, the goal of this study was to investigate 5th grade students' views of explanation, argument, and evidence across three contexts—what scientists do, what happens in science classrooms, and what happens in everyday life. The study also focused on how students' abilities to engage in one practice, argumentation, changed over the school year. Multiple data sources were analyzed: pre- and post-student interviews, videotapes of classroom instruction, and student writing. The results from the beginning of the school year suggest that students' views of explanation, argument, and evidence, varied across the three contexts with students most likely to respond “I don't know” when talking about their science classroom. Students had resources to draw from both in their everyday knowledge and knowledge of scientists, but were unclear how to use those resources in their science classroom. Students' understandings of explanation, argument, and evidence for scientists and for science class changed over the course of the school year, while their everyday meanings remained more constant. This suggests that instruction can support students in developing stronger understanding of these scientific practices, while still maintaining distinct understandings for their everyday lives. Finally, the students wrote stronger scientific arguments by the end of the school year in terms of the structure of an argument, though the accuracy, appropriateness, and sufficiency of the arguments varied depending on the specific learning or assessment task. This indicates that elementary students are able to write scientific arguments, yet they need support to apply this practice to new and more complex contexts and content areas. © 2011 Wiley Periodicals, Inc. *J Res Sci Teach* 48: 793–823, 2011

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The National Research Council (Duschl, Schweingruber, & Shouse, 2007) report, *Taking Science to School*, provides a new framework for proficiency in elementary science, which includes a focus on students being able to “generate and evaluate scientific evidence and explanations” and “participate productively in scientific practices and discourses” (p. 2). Science is a practice that includes more than just concepts and facts, but rather also encompasses scientific ways of thinking and reasoning (Lehrer & Schauble, 2006). In order to

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succeed in science, students need to be able to navigate different discourses or ways of knowing, doing, talking, reading, and writing (Moje, Collazo, Carrillo, & Marx, 2001).

Argumentation is a central scientific practice in the discourse of science in that scientists socially construct knowledge through evaluating scientific claims, weighing evidence, and assessing alternative explanations (Driver, Newton, & Osborne, 2000). Engaging students in scientific argumentation has many potential benefits such as learning science concepts, engaging in scientific discourse, altering students' views of science, and supporting them in socioscientific decision-making (Erduran & Jiménez-Aleixandre, 2008). Young children are capable of quite sophisticated scientific thinking and reasoning, which has been historically underestimated (Duschl et al., 2007). When provided with the appropriate opportunities and support, even elementary students can move beyond simply observing and describing to negotiate and debate meanings and explanations (Varelas, Pappas, Kane, & Arsenault, 2008). One important strategy is to consider students' everyday ways of knowing as a resource to support them in engaging in scientific practices (Warren, Ballenger, Ogonowski, Rosebery, & Hudicourt-Barnes, 2001). Yet additional research needs to be conducted to understand students' views of explanation, argumentation, and evidence across various contexts. Consequently, this study investigated the following research questions: What are elementary students' ideas about explanation, argument, and evidence in the context of science, science class, and their everyday lives? How do elementary students' ideas and their ability to write scientific arguments change over the school year?

Theoretical Framework

Explanation, Argument, and Evidence

Teaching students about science requires engaging them in the disciplinary practice of science with a focus on constructing reliable knowledge claims that provide explanatory accounts of nature (Ford & Forman, 2006). The construction of knowledge claims in science occurs through argumentation in which scientists debate and justify claims using evidence (Driver et al., 2000). This social construction of knowledge is essential to the scientific community in which claims are disputed and change over time as new evidence arises and different theories are debated.

Schools often prioritize final form science instead of engaging students in science as a practice in which students engage in scientific ways of thinking and reasoning (Lehrer & Schauble, 2006). Developmental constraints have been inappropriately cited as reasons why elementary students should not engage in scientific inquiry and reasoning in science (Metz, 1995). Yet recent research suggests that even young students can engage in these essential scientific practices (Duschl et al., 2007; Herrenkohl & Guerra, 1998; Varelas et al., 2008). However, elementary teachers may not give priority to these scientific practices in their teaching, because they do not see them as supporting students' development of content knowledge and they do not see them as educational goals in their own right (Beyer & Davis, 2008). Zembal-Saul (2009) argues for the importance of moving elementary science along a continuum to go beyond just focusing on fun hands-on activities. Elementary science needs to engage students in inquiry in which students support claims with evidence, construct arguments, and consider alternative explanations.

In order for k-8 students to develop proficiency in science they should be engaging in using evidence, generating explanations, and participating in argumentation (Duschl et al., 2007). Although these scientific practices are often cited as important, the distinctions between these practices can be unclear and science educators define them in different ways

(Osborne & Patterson, 2011). I define an explanation in science as providing an account of how or why a phenomenon occurs and explaining why the natural world works in particular ways (Berland & Reiser, 2009; Chinn & Brown, 2000; McNeill, Lizotte, Krajcik, & Marx, 2006). Berland and Reiser (2009) argue that in terms of explanation as a practice of science, it is important for students to understand and participate in how explanations are constructed, questioned, evaluated, and revised through the practice of argumentation. They further describe argumentation as when, “Individuals compare conflicting explanations with the support for those explanations and work to identify/construct an explanation that best fits the available evidence and logic” (p. 28). As such, explanation and argumentation are complementary practices.

Argumentation can be defined in terms of both an individual or structural meaning as well as a social or dialogic meaning (Jiménez-Aleixandre & Erduran, 2008). In terms of the structural meaning or argument product many science educators adapt Toulmin’s (1958) model of argumentation in that a claim or explanation is justified using various supports such as evidence, logic, warrants, and reasoning (see Sampson & Clark, 2008 for a review). The dialogic or social perspective on argumentation focuses on the interactions between two or more individuals in which the participants try to persuade or convince each other of the validity of their claims. Consequently, engaging in argumentation includes the construction and critique of multiple explanations and the use of evidence.

Evidence is data, which can consist of either quantitative or qualitative measurements, that is used to answer a question, solve a problem, or to make a decision (Aikenhead, 2005). Tytler et al. (2001) argue that the use of evidence is central to the interactions between the public and science. Individuals need to consider, ask questions about and use evidence when engaged in regulation, policy formation and decision-making that involve science in their everyday lives. For example, using evidence is important for individuals to engage in problem solving or decision-making in everyday circumstances around a science-related matter, such as deciding to have a child immunized (Aikenhead, 2005). Developing an understanding of and ability to use evidence is an essential component of argumentation and an important practice of the discourse of science for all individuals.

Since the late 1980s there has been an increase in argument-based interventions exploring how to better support k-12 students (Cavagnetto, 2010). These interventions have focused on a variety of different strategies such as the use of curriculum materials (Berland & Reiser, 2009; McNeill et al., 2006), technology tools (Clark & Sampson, 2007; Schwarz, Neuman, Gil, & Ilya, 2003), and teaching strategies (McNeill, 2009; McNeill & Krajcik, 2008; Simon, Erduran, & Osborne, 2006). In his review of scientific argumentation interventions, Cavagnetto (2010) characterizes the studies based on their orientations towards designing learning environments: (1) immersion in practice; (2) explicit instruction in argument; and (3) an understanding of socioscientific issues. Clark and Sampson’s (2007, 2008) work investigating the role of a computer software program in supporting students in argument illustrates an intervention focused on immersion in that student learning occurred by using scientific argument in investigative context. In terms of explicit instruction, Venville and Dawson (2010) found that explicitly teaching scientific argumentation through modeling and other instructional practices as well as providing students opportunities to engage in argumentation supported students in developing stronger argumentation skills as well as conceptual knowledge. Other researchers, such as the work of Sadler and Donnelly (2006), focus on designing interventions around socioscientific issues that emphasize the moral, ethical, and political influences on decision-making in science. Although there has been considerable work in this area, much of the research suggests that students continue to have difficulty with

these practices. Two other reviews of the literature found that students often have difficulty justifying claims with evidence and evaluating arguments (Sadler, 2004) and that students struggle with constructing arguments and evaluating claims (Driver et al., 2000). Research on supporting students in argumentation is in its infancy and we continue to need more work on how to design learning environments to better support student learning (Osborne, 2010).

In designing learning environments and providing other supports for k-12 classrooms, there is a tension in the field around the language choice for these practices particularly in terms of the words “explanation” and “argument”. Osborne and Patterson (2011) argue that the term “explanation” is often inappropriately used in standards documents and in research literature when the actual practice being described is one of argument in which a claim is justified using evidence. In my own previous work with colleagues (McNeill, 2009; McNeill & Krajcik, 2008; McNeill et al., 2006), we struggled with the tension of whether to use the word explanation or argument when working with students and teachers. In this previous work, our goal for students was to write scientific explanations about phenomena (i.e., explain how or why a phenomenon occurs) in which they justified their claims with appropriate evidence and reasoning. As Osborne and Patterson argue the structure of this student writing was consistent with an argument. However, we chose to call the practice a scientific explanation because, “First, the middle school science teachers needed to address the state and national standards that included a focus on explanation... Second, we were concerned about the negative everyday meanings around the term argument that students could bring to the classroom” (McNeill, 2009, p. 236). Osborne and Patterson acknowledge the second issue around individuals’ negative views of the word argument, but then make the case that it is important to change these views. Students’ ideas or beliefs about how knowledge claims are constructed in science class may impact their ability to engage in these practices (Sandoval, 2005). Consequently, if we want to support students in engaging in these practices, we need to consider their ideas or views of these practices. Yet, the majority of the research around argumentation has focused on students’ abilities to engage in this practice, regardless of whether the practice is referred to as explanation or argument, and not on students’ ideas or views of these different terms.

This study focused on examining elementary students’ views of explanation, argumentation, and evidence as well as examining how students’ abilities to construct scientific arguments changed over the school year. The goal of the learning environment was to support students in justifying their claims about how or why the natural world works through using evidence and reasoning, considering multiple alternative explanations, and building on and critiquing the explanations of their classmates.

Science and Everyday Discourses

In considering students’ views of explanation, argumentation, and evidence, it is important to take into account their ideas about these terms across multiple contexts including their everyday lives. Students’ cultural and linguistic backgrounds impact the knowledge and experience that they bring to the classroom, which influences their degree of comfort with the norms of scientific practice (Duschl et al., 2007). Success in school is impacted by students’ ability to acquire academic social language, which can have different norms compared to their “lifeworld” or everyday discourse (Gee, 2005). Students’ views or understandings of these practices may be different in their everyday lives compared to science, yet still impact their ability to successfully engage in these practices in the science classroom.

Little research has directly observed students’ science-related discourse and practice in non-school settings (Bell et al., 2006). Research in this area suggests that children can hold

multiple meanings of content and practices as they navigate their everyday lives. For example, Reeve and Bell (2009) found in their ethnographic study of 13 children aged 9–11 that children held a wide range of meanings for “healthy” across the contexts of their everyday encounters with science and technology. These meanings included the absence of illness, the ability to do desired things (e.g., run fast), emotional health, environmental health, organic foods, and cleanliness. This suggests that students can have different views of terms, like healthy, across different contexts.

Warren et al. (2001) argue for the importance of taking into consideration the continuities between students’ everyday and scientific ways of knowing and talking. By continuities, they suggest that the relationships are more complex than just similarities and differences, but need to take into consideration the depth and complexity of students’ sense-making in science. Their perspective is in contrast with a discontinuous tradition, which views students’ everyday knowledge as incompatible with science. In this work, I take a similar perspective in that I do not view students’ everyday funds of knowledge and practices as a barrier to learning science. Rather, I draw from the work of Moje et al. (2004) who argue for the importance of developing “third space” in science classrooms, which actively integrates the multiple funds of knowledge and discourse that students encounter in both their everyday lives and in more formalized institutions such as school. This third space is a hybrid space based on the idea that individuals in any given context draw on multiple resources. Moje argues that developing third space in science classrooms can serve as a productive scaffold for supporting science literacy as well as can challenge and expand the literacy practices that are typically valued in schools.

Students need to be introduced to scientific discourse in ways that allow them to bridge their everyday discourse and experiences with the academic language (Varelas et al., 2008). In classrooms, science discourse can present a challenge for students and can serve as a gatekeeper preventing access to the culture of science (Brown, 2006). Educators can use the varied repertoires of practice students develop in their everyday lives in order to support them with academic disciplinary practices (Nasir, Rosebery, Warren, & Lee, 2006). Classroom instruction should be informed by students’ everyday ways of knowing, which can be a generative practice for developing science discourse in elementary classrooms (Warren et al., 2001). In addition, making the language of science explicit can support students in engaging in scientific discourse and in developing conceptual understandings (Brown & Spang, 2008). In terms of reading and writing, schools often focus on the personal narrative with limited exposure to more academic genres, such as scientific argumentation, which can be challenging for all students particular those not exposed to more academic discourses in their homes (Purcell-Gates, Duke, & Martineau, 2007). Making visible the structure of the domain, building from students’ everyday ways of knowing, actively engaging students in academic discourse and providing timely and flexible feedback can help support students in developing expertise in science (Nasir et al., 2006). Engaging in these types of strategies to develop third space requires that teachers, curriculum developers, and other educators develop rich understandings of students’ everyday funds of knowledge and discourses (Moje et al., 2004).

Currently, we need more research about students’ views and abilities to engage in the practices of explanation, argumentation, and evidence in their everyday lives. Research in this area suggests that some practices do seem to have similar meanings in everyday and scientific contexts, such as the use of evidence (Bricker & Bell, 2007), while other practices can have different meanings, such as students’ viewing argument as yelling or fighting (Bricker & Bell, 2008). Other research on argumentation has found more continuity between everyday and scientific argumentation. In Hudicourt-Barnes’ (2003) work with elementary and middle

school students, she found that the Haitian cultural practice of *bay odyans* is a similar form of discourse to scientific argumentation. This common form of talk in Haitian culture includes debate in which participants discuss divergent points of view and support those viewpoints using evidence or logic. These debates can also be quite theatrical including funny interjections and laughter, which differs from scientific argumentation, yet this everyday practice provides students with many resources they can draw upon to engage successfully in scientific discourse. Hudicourt-Barnes stresses the importance of incorporating students’ existing everyday practices into science teaching. In previous work with my colleagues (McNeill & Pimentel, 2010), we found that students also used everyday knowledge in their scientific practices. When students engaged in argumentation around the socioscientific context of global climate change, students frequently drew from their personal experiences and other experiences outside of their science classroom to use as evidence to support their claims. Although students made connections across their everyday lives and science classrooms, it was unclear if they viewed the use of evidence or the practice of argumentation as different in these two contexts. Consequently, this study focused on investigating students’ views about evidence, explanation and argument in science, the science classroom, and in their everyday lives as well as how to best support them in successfully engaging in scientific argumentation.

Methods

Context of the Study

This study took place in an elementary school in a large urban school district in New England with a grades 3–5 elementary science specialist, Mr. Cardone, and his 5th grade students. Mr. Cardone was certified to teach elementary education and general science for grades 5–8; furthermore, he had been teaching elementary science for 6 years at the time of the study. The student body at the school was ethnically diverse with approximately 15% African American, 60% Latino/a, 12% White, 12% Asian, and 1% Other. Approximately 82% of the students attending the school were eligible for free or reduced lunch. Mr. Cardone taught two sections of 5th grade with approximately 18 students in each class. The specific number of students varied over the school year, because of student mobility in and out of the school district. Table 1 provides a summary of the students’ backgrounds in Mr. Cardone’s two 5th grade classrooms. The table includes 33 of the 36 students from the beginning of the school year who completed the background information sheet.

As the table illustrates, Mr. Cardone’s 5th grade classes were very diverse with the majority of students speaking another language in addition to English.

Table 1
Student backgrounds (N = 33)

Gender	Race	Birth Country	Speak Another Language	What Language
Male: 15	Black: 3	USA: 22	English only: 6	Bangia: 1
Female: 18	Asian: 7	Aruda/Bonire: 1	Other language: 27	Cambodian: 2
	White: 3	Colombia: 1		Creole: 1
	Latino/Latina: 17	Dominican Republic: 2		Portuguese: 2
	Multiracial: 3	El Salvador: 3		Spanish: 20
		Guatemala: 3		Vietnamese: 1
		Haiti: 1		

Study Design

To investigate the research questions, Mr. Cardone and I engaged in design-based research (Brown, 1992) during the 2008–2009 school year. Design-based research is important for understanding the how, when and why educational practices work in classroom contexts (The Design-Based Research Collective, 2003). Design-based research focuses on understanding the messiness of real-world learning environments through the collection and analysis of multiple data sources and the flexible design and revision of the learning environments (Barab & Squire, 2004). Typically, design-based research includes the close study of a single learning environment as multiple iterations of instructional strategies are tested to investigate their relationship with learners' beliefs and ideas. Those multiple iterations can occur within a single curriculum or single school year as a variety of different instructional strategies are explored. In contrast to traditional experiments that include control groups, design-based research tries to develop rich characterizations of both the learning environment and student learning (Puntambekar, Stylianou, & Goldstein, 2007). This type of methodology can be appropriate when exploring an innovative area of research in which not all variables of interest are known in advance and there is a complex relationship between the variables, which require procedures that are flexible and can evolve through the study (Barab, 2006). The research has the dual goals of impacting educational theory and classroom practice (Collins, Joseph, & Bielaczyc, 2004). As Barab and Squire (2004) argue, "Design-based research requires more than simply showing a particular design works but demands that the researcher (move beyond a particular design exemplar to) generate evidence-based claims about learning that address contemporary theoretical issues and further the theoretical knowledge of the field" (p. 5–6). Design-based research supports the development of new theories and practices, which can then be later tested through more controlled experimentation. Specifically in this study, we were interested in designing an elementary classroom environment that supported students in argumentation, but we were also interested in investigating how the students' views and abilities to engage in argumentation change over the course of the school year.

Multiple data sources were collected to examine students' views of explanation, argument, and evidence as well as their ability to engage in scientific argument. Data sources included: student pre- and post-interviews, student writing for the lessons focused on argumentation, student writing from a district wide common writing assignment, student writing from end of the year assessments, and videos tapes of the seven lessons focused on argumentation. Furthermore, the multiple data sources and analysis supported the design-based research in that they enabled us to systematically engineer the context to investigate additional instructional strategies in this real-world learning environment (Barab, 2006).

For the pre- and post-interviews, 23 students were interviewed during the second week of school and then again during the last month of the school year. All students were interviewed who returned their consent forms providing permission before the pre-interview. During the week before the interviews, Mr. Cardone did not explicitly talk about explanation, argument or evidence during science class. The interview focused on students' views of these three terms (i.e., explanation, argument, and evidence) in three different contexts—what scientists do, what happens in science classrooms, and what happens in everyday life (e.g., home or with friends). Table S1 provides the interview protocol that was used at both the beginning and end of the school year. During the interview, if the student brought up a term or an important concept without explaining that idea, the interviewer followed up with the prompts at the end of the protocol. For example, if a student mentioned that explanations in science

use data, the interviewer might follow up with—You mentioned data. What do you mean by data? Originally, 13 students were interviewed in class A and 11 students were interviewed in class B (i.e., 24 students total). One of the interviewed students in class B moved during the school year and we were unable to conduct a post-interview. Consequently 23 students were interviewed at both the beginning and end of the school year.

In accordance with design-based research methodologies (Barab & Squire, 2004), the design process was ongoing throughout the year as we continued to design lessons, instructional strategies, and curricular scaffolds informed by the student interviews, classroom videotapes, and student writing as well as the current research literature. Each lesson that we developed included an iterative cycle of implementation, assessment, and refinement to inform future lessons. Mr. Cardone was scheduled to see each class three times a week for an hour at a time, though sometimes he saw his students less frequently because of fieldtrips, holidays, or other special events. At the beginning of the year our intention was to design one lesson focused on scientific argumentation for each month. Two months we were unable to include a lesson, because of time restrictions (February and April). We designed a total of seven lessons throughout the school year, which incorporated scientific argumentation (Table S2). Six of the 7 lessons asked students to construct a scientific argument for one specific investigation that they completed in class. The one exception to this is Lesson 5 when students were provided with a more open prompt in which they were asked to write a scientific argument using any of the data and information they had collected during the multiple investigations in the levers and pulleys unit. In addition, 6 of the 7 lessons asked students to write an original scientific argument with the exception of Lesson 6 in which students selected the strongest claim, evidence, and reasoning from multiple options. This unique lesson is described in more detail in the Results Section (Figure 1). There were also two additional scientific argumentation writing tasks included in the dataset. One writing task was a district wide common writing assignment, which was a collaboration between the language arts teacher and the science teacher. This assignment was completed between Lessons 6 and 7 and the students developed drafts and received feedback before writing their final arguments. The second writing task was a question included on the end of the year assessment that student answered independently and without feedback. Table S2 provides a summary of the seven lessons and the two writing tasks including the specific writing prompt for each assignment.

Based on preliminary analyses of the beginning of the year student interviews and previous research supporting middle school students in scientific explanation and argumentation (McNeill, 2009; McNeill & Krajcik, 2008), Mr. Cardone and I designed Lesson 1: Habitats to introduce his students to scientific argumentation. We used a framework that consists of three components (McNeill et al., 2006): (1) Claim—a statement that answers a question or problem; (2) Evidence—scientific data that supports the claim. The data needs to be appropriate and you need to have enough data; and (3) Reasoning—applies the science knowledge to solve the problem and explains why the evidence supports the claim. This initial lesson lasted 2 days. During the first day of the lesson, students went to the outdoor classroom in their schoolyard and collected data to answer the question: How many different habitats are in the outdoor classroom? After collecting the data, students were asked to write an argument that answered the question. On the second day of the lesson, Mr. Cardone used an everyday example—How long should recess be?—to have a discussion with his students about how to construct a strong argument. At the end of the discussion, Mr. Cardone introduced the framework of claim, evidence, and reasoning using the definitions provided above. He then asked his students to revise the arguments they wrote the day before using this framework.

Mr. Cardone enacted all of the lessons with both of his 5th grade classrooms. One of the two classrooms was selected as the focal classroom for the design-based research, because all of the students handed in their permission slips allowing them to be videotaped. This classroom was videotaped and the student writing was collected after each lesson.

Data Analysis

The coding schemes for the student interviews, student writing, and videotapes of the seven lessons were developed from the theoretical framework and iterative analyses of the data sources (Miles & Huberman, 1994). The coding schemes were developed to inform the overarching research questions examining elementary students' ideas about explanation, argument, and evidence in the context of science, science class and their everyday lives, and how their ideas and the ability to write scientific arguments changed over time.

Student Interviews. The analysis of the student interviews focused on students' ideas about evidence, explanation, and argumentation in the different contexts. All student interviews were transcribed and the transcripts were used to develop the coding schemes and to code the data. The coding scheme was informed by previous research examining students' everyday meanings of evidence, explanation, and argumentation (Bricker & Bell, 2008; Hudicourt-Barnes, 2003) and previous research examining scientific evidence (Aikenhead, 2005), explanation (Berland & Reiser, 2009; McNeill et al., 2006), and argument (Osborne, Erduran, & Simon, 2004; Sampson & Clark, 2008). For example, two of the codes for evidence included "data" and "support an answer to a question", based on Aikenhead's (2005) discussion of these as key characteristics of evidence. One of the codes for explanation included "how or why a phenomenon occurs," since this is how the term has often been defined in the science education literature (Berland & Reiser, 2009; McNeill et al., 2006). The codes for argument included "persuade or convince," because this definition is found in both the everyday practices literature (Hudicourt-Barnes, 2003) as well as the science education literature (Jiménez-Aleixandre & Erduran, 2008). The argument codes also included "emotional or angry fighting," because this concept has also been associated with argumentation in students' everyday lives (Bricker & Bell, 2008). The development of the coding scheme for the interviews was also grounded in the students' own language (Strauss, 1987). In examining the transcripts, we found that students brought up ideas, which we had not previously identified as potential codes from the literature. For example, we found that the idea of an "exchange between people" often came up in students' comments when discussing all three terms. Consequently, we added this code to evidence, explanation, and argument. In contrast, students discussed the idea of "how or why a person does something" only when talking about explanations so the code was added to that coding scheme.

The results include the specific codes (Tables 7–9) as well as examples to illustrate the most common codes during the interviews. Students' responses in the interview could receive multiple codes if they discussed different ideas in relation to one of the practices. For example, in discussing evidence a student could talk about both "data" and "support an answer to a question" so their response was coded for both ideas. Two independent raters, the author and a research assistant, coded the interviews. Inter-rater reliability was calculated by percent agreement. The two raters overlapped on 14 of the 46 interviews or 30%. The percent agreement for the interviews was 84%. All disagreements were resolved through discussion. We were interested in how the students described the terms in the three different contexts as well as how those definitions changed over the course of the school year. Consequently, we

summed the number of students who received each code and calculated the percentage of students to illustrate how the ideas of the class as a whole either did or did not change.

Student Writing. The student writing was coded using two different coding schemes: genre and argument structure. The focus on genre stemmed from the work of Purcell-Gates et al. (2007), who found that in terms of reading and writing schools often focus on the personal narrative with limited exposure to more academic genres, such as scientific argumentation. The genre-coding scheme focused on whether the general structure was that of an argument or aligned more closely with another writing genre. The writing style was coded regardless of the accuracy of the science content using four different codes: 1—Argument (a claim supported with evidence and/or reasoning); 2—Just claim (a claim with no justification); 3—Informational text (reporting information about the science, but no claim is provided); and 4—Personal narrative (a story about the students’ thoughts, feelings, or actions). The genre-coding scheme captured the overall structure of the student writing. Table 2 provides the four codes, a general description, and specific student examples from Lesson 1 to illustrate the student writing for each code. In Lesson 1, students wrote scientific arguments that addressed the question: How many different habitats are in the outdoor classroom? The argument example illustrates a student who provided a claim that there were three different habitats (i.e., dirt and underground, tree, and log/rocks) and then continues on to justify why she felt each of those examples is a habitat. Because of the structure of including a claim and then justifying the claim, the student’s writing was coded as an argument. This structure differs from the informational text in which the student is just describing a spider or the story in which the student is describing her experience observing the eggs.

Each of the eight samples of student writing from across the school year was coded for the one genre type that was most prevalent in the writing sample. We summed the number of students who received each genre at each time point. We then calculated the percentage of students in the class that received the genre in order to illustrate any changes in writing as a class over time.

Table 2
Genre coding scheme with examples from lesson 1

Code	Description	Student Examples—Lesson 1
Argument	Justifies a claim using either evidence or reasoning. The writing needs to have a claim and some justification for that claim	“Dirt and underground, tree, log/rocks. I choose #1 because worms live in the ground and dirt. And food for them they go up ground and search on flower and leaves. I choose #2 because. . .”
Just a claim	Provides a claim with no justification (i.e., evidence or reasoning). A claim is a statement that answers the question	“Both of them are habitats—dirt and under rocks”
Informational text	Reports information about the science content, but does not provide a claim for the specific question asked. This could include a scientific definition or a description of scientific observations	“The spider has eight legs. And it could make a web to catch his food which it is fly”
Story	Includes a story about thoughts, feelings, or actions. This could talk about a student liking to do something or a creative story about a plant or animal	“Those eggs really look weird and was my first time looking at eggs”

For the argument structure-coding scheme, similar to other science educators (Sampson & Clark, 2008), we adapted Toulmin's model of argumentation (1958) to code student writing in terms of claim, evidence, and reasoning (McNeill, 2009; McNeill et al., 2006). The claim is the statement or conclusion that answers the original question or problem. Claims were coded as either 0 or 1 for either appropriate or inappropriate. The evidence is scientific data that supports the claim, which needs to be both appropriate (scientifically acceptable for the claim) and sufficient (include enough pieces of data to support the claim). Evidence was coded as 0 for inappropriate or missing, 1 for appropriate and insufficient, and 2 for appropriate and sufficient. Reasoning is a justification that connects the evidence to the claim and shows how or why the data counts as evidence by using appropriate and sufficient scientific principles. Reasoning was coded as 0 for inappropriate or missing, 1 for appropriate and insufficient, and 2 for appropriate and sufficient. Table 3 provides the codes for claim, evidence, and reasoning, a general description of each code, and specific examples from Lesson 3 in which students wrote arguments to answer the question—Is my ecocolumn a stable ecosystem? A specific rubric for each of the eight writing assignments was developed to clarify what counted as appropriate and sufficient for each content area and writing task.

Since the codes for evidence and reasoning included a range of quality (levels 0–2), we calculated the average student score for each time point. We then graphed how the average score changed over time (Figure 3). This is in contrast to our data reduction for the interviews and the coding of the genre of writing, because those coding schemes focused on the presence or absence of particular codes (coded 1 or 0), which enabled us to examine the percentage of students who received each code over time.

Similar to the interviews, the student writing was coded by two independent raters. Inter-rater reliability was calculated by percent agreement. For the student writing, both independent raters coded all students writing. The percent agreement for the writing was 84%. All disagreements were resolved through discussion.

Videotapes. Finally, the seven videos were coded in order to provide an overarching context of the learning environment in which the students were being supported in constructing scientific arguments. For each video, a narrative description was created with time stamps provided throughout so the video could be referenced for clarification if needed. The narratives provided summaries for aspects of the lesson not focused on argumentation (e.g., Lesson 3 defining stable vs. unstable ecosystems) and detailed descriptions of the parts of the lesson focused on argumentation including specific quotes from Mr. Cardone and his students to illustrate the discussions. The narratives were then coded for instructional strategies that stemmed from previous research examining the role of the teacher in supporting students in scientific argumentation (McNeill, 2009; McNeill & Pimentel, 2010; Simon et al., 2006). Each lesson was coded for the presence of six different instructional strategies: (1) discuss a framework for scientific argument (claim, evidence, and reasoning); (2) model and critique examples; (3) connect to everyday arguments; (4) provide students with feedback on their arguments; (5) have students engage in peer critique; and (6) debate arguments as a whole class. Two independent raters coded each narrative description for the six instructional strategies. Again, the inter-rater reliability was calculated by percent agreement with 88% agreement.

The analysis also included member check in that Mr. Cardone was an active participant in the research process. During the school year as the coding schemes for the student interviews and student writing were developed, he provided feedback on whether he felt the

Table 3
Argumentation coding scheme with examples from lesson 3—Is my ecocolumn a stable ecosystem?

Claim	Evidence	Reasoning
<i>A statement or conclusion that answers the original question/problem</i>	<i>Scientific data that supports the claim. The data needs to be appropriate and sufficient to support the claim</i>	<i>A justification that connects the evidence to the claim. It shows why the data counts as evidence by using appropriate and sufficient scientific principles</i>
0 Does not make a claim or makes an inappropriate claim <i>Lesson 3—Stable Ecosystem:</i> “Snails live in water”	Does not provide evidence or only provides inappropriate evidence or vague evidence <i>Lesson 3—Stable Ecosystem:</i> “Because the plant gave food”	Does not provide reasoning, or only provides inappropriate reasoning <i>Lesson 3—Stable Ecosystem:</i> “Because I took the top off and smelled it and it wasn’t that smelly”
1 Makes an appropriate and scientifically accurate claim <i>Lesson 3—Stable Ecosystem:</i> Two potentially appropriate claims— The ecocolumn is stable or the ecocolumn is not stable depending on the student’s data “My ecocolumn isn’t stable” “My ecocolumn is stable”	Provides one piece of appropriate evidence, but the evidence is insufficient. The evidence needs to align with the student’s claim <i>Lesson 3—Stable Ecosystem:</i> Evidence may include data about plants, fish, snails, pill bugs, or worms in terms of health, growth, or numbers OR could include water quality data “My evidence is that the snail is having snail eggs it is moving a lot”	Provides appropriate, but insufficient reasoning. Often does not include a complete explanation of scientific principles to explain how or why the evidence supports the claim <i>Lesson 3—Stable Ecosystem:</i> Describes the characteristics of a stable or unstable ecosystem (e.g., a stable ecosystem has living plants and animals), but does not include the idea of organisms being able to survive and reproduce over time “Stable ecosystems have living animals, plants growing, living insects, and clean water”
2 Not applicable	Provides 2 or more pieces of appropriate evidence to sufficiently support the claim. The evidence needs to align with the student’s claim <i>Lesson 3—Stable Ecosystem:</i> Evidence may include data about plants, fish, snails, pill bugs, or worms in terms of health, growth, or numbers OR could include water quality data “First of all our bugs died. The water doesn’t look clear and Lastly the plants stopped growing!”	Provides appropriate and sufficient reasoning including scientific principles explaining how or why the evidence supports the claim <i>Lesson 3—Stable Ecosystem:</i> Explains that an ecosystem is stable if all of the organisms are able to survive and reproduce over time. Explicitly connects this scientific principle to explain why the evidence supports the claim “In stable ecosystems, plants and animals grow, reproduce and die over time. Some of my plants and insects died, but others grew and had babies so over time it is stable”

codes encompassed his students’ ideas and abilities about scientific argumentation. The teacher instructional practices for the coding of the videos were all strategies that were discussed and developed with Mr. Cardone as part of the design-based research. After writing up the research, Mr. Cardone also read the results to provide feedback. He felt the results

adequately represented his classroom in terms of his practices and the ideas and abilities of his students.

Results

The analyses address the following two research questions: (1) What are students' ideas about the terms explanation, argument and evidence in the context of science, science class and their everyday lives and how do they change over the school year? and (2) How do students' abilities to write scientific arguments change over the school year? Before addressing these research questions, I begin by describing the teacher instructional strategies used throughout the school year to provide the contextual background that helps explain the change in students' ideas and abilities over time.

Teacher Instructional Strategies

Throughout all seven lessons, Mr. Cardone consistently provided his students with support for scientific argumentation using a variety of different instructional strategies. Table 4 provides an overview of the lessons and the instructional strategies used in each lesson. Mr. Cardone used each of the six instructional strategies in at least one lesson and used one of the instructional strategies (i.e., Provide students with feedback) in six of the lessons. In terms of connecting to everyday arguments, this strategy occurred in two distinct lessons. In Lesson 1, he used the example of the length of recess to introduce the claim, evidence, and reasoning framework while in Lesson 4 he used an everyday example about skiing to again discuss similarities and differences between the everyday practice of constructing an argument and constructing an argument in science class.

The specific strategies he used shifted some over the course of the school year. As Table 4 illustrates, in the beginning of the school year Mr. Cardone was more likely to spend time actually discussing the framework and defining the different components. Lesson 5 is the last time he explicitly defined the different components of the argumentation framework

Table 4
Instructional strategies in seven argument lessons

	Discuss Framework	Model and Critique Examples	Connect to Everyday Arguments	Provide Students With Feedback	Have Students Engage in Peer Critique	Debate Arguments as a Whole Class
Lesson 1: Habitats	X	X	X	X		
Lesson 2: Adaptations	X	X		X		
Lesson 3: Stability of Ecosystems	X	X		X	X	
Lesson 4: Levers	X		X	X		
Lesson 5: Levers and Pulleys	X			X		
Lesson 6: Force and Motion						X
Lesson 7: Pendulums				X		

and here he was just reviewing ideas he had already discussed with his students. For example, when introducing the levers and pulley activity in this lesson he said:

Now again, just to review the process—the first step is to be making your claim. So you're going to look through the work that we've done together, in your notebooks, in the handouts that you have from levers and pulleys and you're going to come up with a claim that you have evidence for in your notes. Okay? So again, a claim is just a statement that you're making. Step two is to provide the evidence for your claim. So here you're going to be using your notes to find that evidence and this is going to be backing up the claim that you state. Okay? Any questions on that? Evidence. (Pauses). And then the last part, step three—we're going to be providing the reasoning. This part explains how your evidence supports your claim. Okay? So, any questions on the reasoning piece? Okay, it's all stuff that we've been done before and every time we do it you guys are getting better and better at it.

The later lessons focused more on having students create scientific arguments and providing them feedback on their arguments.

Mr. Cardone also tried a new strategy in Lesson 6 in which he had students debate the quality and strength of their claims, evidence, and reasoning for an argument addressing the question: How do you design a car to go the fastest? In this lesson, the students worked in groups to select from a student sheet which 1 of 3 claims was the strongest, which 2 of 6 pieces of evidence was the strongest to support that claim, and which 1 of 3 reasoning statements best explained why the evidence supported the claim. The choices aligned with the data the students had previously collected during investigations exploring how load and force impact the speed of a car. For example, Figure 1 includes the three different reasoning choices from which the students had to select. After the groups made their choices, they then debated as a class which choices were the strongest. In this lesson, the focus was on helping students develop a stronger understanding of what counted as strong evidence and reasoning. For example, in the case of the reasoning, Mr. Cardone's goal was to help the students understand that choice C was the strongest choice, because it included an explanation of speed, which was an important science concept that they had to apply in order determine which evidence best supported the claim about the fastest car. He wanted to help students understand that reasoning should include a scientific concept that articulates why the evidence supports the claim.

Mr. Cardone started by having each group quickly state which reasoning statement they selected to see if there was disagreement in the class. One group selected B and the other groups were split between A and C. He then had the students debate as a class, which one

REASONING

Circle ONE of the following.

- A. The data from our experiments shows us how to build our car. Since the data shows that fast cars have a light load and fast cars are pulled by a large force then this is how we should build our car.
- B. Since car companies and race cars have cars that are really light and have large engines this means we should design our car in the same way. It should have a light load and be pulled by a large force.
- C. The speed was determined by how many seconds it took for the car to travel across the table. The car with less blocks had a lighter load and it traveled faster. The car that was pulled by more washers was pulled by a greater force and it traveled faster.

Figure 1. Three reasoning choices for argument debate.

they thought was strongest and why. Table 5 includes an excerpt from the beginning of the discussion.

This transcript illustrates that by the end of the year the students were not asking Mr. Cardone what he meant by a scientific argument, rather they were debating how to construct the strongest argument. The students debated what counted as the strongest evidence or reasoning for a particular investigation as they applied the framework to try and make sense of a particular dataset. They had developed an understanding as a class that scientific arguments consist of claims, evidence, and reasoning, and now were focusing on understanding what that meant in different contexts.

Students' Ideas about Explanation, Argument, and Evidence

The analysis of the interviews revealed that students' views of explanation, argument, and evidence varied across the three different contexts both at the beginning and end of the school year. Furthermore, students' views of the meanings of the terms for scientists and for science class changed over the course of the school year, while their everyday meanings remained more constant. In this section, I present the codes for each of the practices as well as quotes to illustrate the most common student ideas for each of the three contexts. As such the findings focus on the ideas of the majority of the students, rather than representing all of the potential codes used in the analysis. In selecting the quotes, I balanced picking the quote that most clearly illustrated the code with selecting a variety of different students to include

Table 5
Class debate about reasoning

Participant	Transcript
Mr. Cardone	All right I'm going to let Lucia and Kiyana start us off. All right
Kiyana	We chose A
Lucia	I got C
Mr. Cardone	Okay. Kiyana why did you think A? Hold on a second. Randall, everybody needs to now be listening to Kiyana. No, don't. We're done explaining, I need you to listen now, cause we're going to hear each others arguments, I appreciate your comment but you guys disagree, so you're going to convince the whole class not just her. That's why I don't want you to share with her right now. All right, Kiyana go ahead
Kiyana	They chose B but they weren't correct because um in A it's explaining A it's explaining [evidence] A and D in reasoning because in [evidence] D it says the car was pulled by 5 washers by heavy force and it took 2 seconds and the other car had one washer and it took a lot of time because its car was really heavy and in reasoning A is explaining [evidence] D and A
Mr. Cardone	She's trying to say that A is. . .
Kiyana	In A it says the data shows that fast cars have a light load and fast cars are pulled by a large force. It's explaining [evidence] A and D
Mr. Cardone	Okay, So, Lucia let's hear from you know you picked C so let's see why
Lucia	Okay it's C because it's the same thing it's also explaining [evidence] A and D because it's saying the speed is determined by how many seconds it took to travel across the table and um, and how um, the car with less blocks had a higher, had a lighter load and it traveled faster, the car that was pulled by more washers was pulled by a greater force and it traveled faster
Kiyana	A and C both make sense. They're basically saying the same thing except one has more information than the other
Mr. Cardone	So which one has more information?
Lucia	C

as many student voices as possible. At the end of the section, I summarize the trends in the students’ ideas (Table 9).

Ideas about Explanation. Table 6 displays students’ views about explanation. The bold numbers highlight the codes that appeared with the greatest frequency in either the pre- or post-interview. The cells in the table include the percentage of the total number of students ($n = 23$) as well as in parenthesis the actual number of students that received each code. Students’ responses could receive multiple codes (e.g., both observation and exchange between people) if their response included multiple ideas. For example, on the pre-interview 34 codes were assigned for the 23 students in terms of their ideas for what it means for a scientist to create an explanation.

When asking the students about scientists, the interview included both asking students what it means for a scientist to create an explanation and asking for an example of an explanation a scientist might create (Table S1). In coding their responses to these two questions, at the beginning of the school year the majority of students talked about an *exchange between people* while at the end of the year their ideas shifted to focus on *how or why a phenomenon occurs*. For example, at the beginning of the year Denzel focused on scientists talking to other people, “like if they tell somebody, like all the people, like in public that they learned something like new.” His response focused on the interaction between people and sharing something new. When asked for a specific example, he still talked more generally about “well what they like, discovered.” At the end of the year, his ideas shifted in that instead of talking about learning something new with other people, he specifically talked about scientific phenomena. Denzel said that, “. . . if you put, if you put a glass of water in the freezer it will freeze after time. . . liquid turns into solid when it freezes.” Here he explained the phenomenon of how liquid becomes a solid. Two interesting shifts in his language include the focus on specific scientific phenomena and the shift from using “they” to “you.” In terms of the focus on scientific phenomena, this was characteristic of many students in Mr. Cardone’s classes. At the end of the year, students tended to use specific examples of natural phenomena to illustrate the types of explanations that scientists create suggesting they had a stronger understanding of what counted as a scientific explanation. Students were less likely to discuss in general that scientists creating explanations means that they explain ideas to other people. At the end of the year, only 17% of the students talked about a scientific explanation as an exchange between individuals. The shift from “they” to “you” was more unique for Denzel. There are many possible reasons for this shift in language. One possibility is that Denzel may

Table 6
Students’ views about explanation (n = 23)

	Scientist		Science Class		Everyday	
	Pre	Post	Pre	Post	Pre	Post
Link to argument	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	4% (1)
Link to evidence	4% (1)	22% (5)	0% (0)	9% (2)	0% (0)	9% (2)
Writing	4% (1)	9% (2)	13% (3)	4% (2)	0% (0)	0% (0)
Observation	48% (11)	26% (6)	39% (9)	13% (3)	35% (8)	9% (2)
Exchange between people	57% (13)	17% (4)	17% (4)	30% (7)	74% (17)	83% (19)
How or why a person does something	4% (1)	9% (2)	0% (0)	13% (3)	30% (7)	70% (16)
How or why phenomenon occurs	26% (6)	70% (16)	13% (3)	52% (12)	4% (1)	4% (1)
Other	4% (1)	4% (1)	9% (2)	4% (1)	4% (1)	0% (0)
Don’t know	0% (0)	4% (1)	43% (10)	17% (4)	13% (3)	4% (1)

have begun to identify himself as a scientist, rather than just seeing scientists as a distant other.

In the pre-interview, students were more likely to say that they *did not know* what it means to create an explanation in science class (43% of the students) compared to in the other two contexts. Students were less likely to offer an initial idea about science class. By the end of the school year, students' responses about explanations in science class shifted to focus more on *how or why a phenomenon* occurs similar to how they responded about scientists. For example, Mariela talked about a scientific explanation she created in the outdoor classroom, "My explanation was that, like there was a lot, a lot of bees going around the flowers because they have pollen and that's why." In this example, she explained why she observed bees near flowers. Again, here the 5th graders tended to use specific examples to illustrate a scientific explanation from their science class that articulated how or why a natural phenomenon occurred.

In contrast, students' ideas about what it means to create an explanation in an everyday context remained more constant throughout the school year. Specifically, students tended to focus on an *exchange between people* both on the pre- and post-interview. For example, Warren actually used the same example at the beginning and end of the school year for this question. At the beginning of the school year he stated, "you're trying to explain like a game to your friends or something" and at the end of the year he responded, "like it's like you're playing a game and then you explain. Like, say you're playing baseball and you think uh, a guy is out cause you tagged him then you explain like if you got him out first like before he touched the base and then you give an explanation." Both responses were coded as an exchange between people, because of the focus on explaining an idea to another person. In addition, his response at the end of the year was also coded as how or why a person does something because of the additional detail about explaining why the person was out during the baseball game. The initial focus on an exchange between people is similar to how students responded about scientists at the beginning of the school year. In both contexts, students were most likely to talk about two or more individuals interacting with each other. Yet while students' ideas about scientists and in the science classroom shifted, they kept a more constant view that explaining in your everyday life had to do with interactions between other people and not how or why natural phenomena occur.

Ideas About Argument. Table 7 provides the results from students' discussions of what it means to create an argument. Across all three contexts, students were most likely to focus on an *exchange between people*. Student responses were further classified into three different sub-codes based on ideas in the literature and students' own language in the transcripts: persuade or convince (Hudicourt-Barnes, 2003; Jiménez-Aleixandre & Erduran, 2008), disagreement (students' own language), and emotional or angry fighting (Bricker & Bell, 2008). As discussed previously, if a student's response included more than one idea (i.e., both persuasion and disagreement), both codes were used. Students tended to focus more on disagreement in all three contexts rather than on persuasion or angry fighting.

When discussing what it means for a scientist to create an argument during both the beginning and end of the school year, students were most likely to talk about a disagreement. For example, during her pre-interview Latisha explained, "if there's two different scientists and they both have different explanations and if a lot of people don't really agree, or a lot of people agree with one and a lot of people agree with the other they start an argument, but it's not like a fist fight but it's an argument saying that I disagree or I agree." In her response, she even explained that an argument is different from a fist fight clarifying that she did not

Table 7
Students' views about argument (n = 23)

	Scientists		Science Class		Everyday	
	Pre	Post	Pre	Post	Pre	Post
Link to explanation	13% (3)	9% (2)	0% (0)	9% (2)	9% (2)	9% (2)
Link to evidence	4% (1)	17% (4)	0% (0)	9% (2)	4% (1)	9% (2)
Writing	4% (1)	9% (2)	4% (1)	4% (1)	9% (2)	0% (0)
Observation	13% (3)	9% (2)	13% (3)	9% (2)	4% (1)	0% (0)
Exchange between people	83% (19)	83% (19)	43% (10)	70% (16)	87% (20)	91% (21)
Persuade or convince	13% (3)	17% (4)	4% (1)	9% (2)	13% (3)	13% (3)
Disagreement	57% (13)	65% (15)	22% (5)	57% (13)	39% (9)	70% (16)
Emotional or angry fighting	22% (5)	9% (2)	13% (3)	9% (2)	35% (8)	22% (5)
One individual thinking through an idea	4% (1)	4% (1)	0% (0)	0% (0)	4% (1)	0% (0)
Other	9% (2)	0% (0)	4% (1)	0% (0)	9% (2)	0% (0)
Don't know	0% (0)	9% (2)	48% (11)	22% (5)	0% (0)	9% (2)

see scientists' arguments as being physical or necessarily emotional. On the post-interview, Latisha again talked about scientists disagreeing, but here she also brought in the idea of using data or evidence in an argument. She stated, "Like when somebody disagrees with them they would list every single thing, detail they had and create an argument like out of their data and stuff and be like this is what I found out and this is why I think this." Her response actually became more sophisticated in her inclusion of the use of data in an argument. Unfortunately, relatively few students brought up the idea of using data or evidence in a scientists' argument during either the beginning or end of the school year.

Similar to explanation, at the beginning of the school year the students were most likely to say that in terms of science class that they did not know the meaning of argument. By the end of the school year, students' views of argument in science class shifted to focus more on an exchange between people. For example, on the post-interview when discussing argument in science class Carlos responded, "You're trying to convince someone to agree with you. Like arguing, argument. Argument when like someone disagrees with you or something, in the claim or something." His response was coded both as persuade or convince and as disagreement, because both ideas were included in his response. Carlos talked about having a disagreement about a claim rather than people being angry or mad at each other. Again, here students' ideas about science class begin to align more closely with their ideas about scientists by the end of the school year.

In terms of emotional or angry fighting, the highest percentage of student responses occurred on the pre-interview in the everyday context, though only 35% of the students mentioned this idea. For example, on the pre-interview Walter stated, "it's like you get into a fight, like get really mad at someone and start fighting with them about something." His response aligns with previous research suggests that students can view argument as fighting in their everyday lives (Bricker & Bell, 2008). Walter's response on the post-interview illustrates a shift in his ideas to focus more on disagreement rather than on emotions or anger. He stated, "Like, to disagree with someone, like, like sometimes my friends want to go to like some places and then I want to go to somewhere else, like we want to go to one park and they're like, we want to go to the other." On the post-interview a larger percentage of students were more likely to talk about disagreement in their everyday lives. Although the focus on an exchange between people did not alter pre to post, the examples they used shifted to highlight a disagreement from their everyday lives. Students' ideas about arguments seemed to be fairly

similar across the contexts of scientists, the science classroom and their everyday lives at the end of the school year focusing on an exchange between people.

Ideas About Evidence. Table 8 displays students' views about evidence. The two most frequent codes for the student responses were *support an answer to a question* and an *exchange between people*.

In terms of scientists, on both the pre-interview and post-interview the majority of students focused on evidence in terms of *support an answer to a question*. For example, on the pre-interview Sofia stated, "like when they need to find out something they have to make sure they have evidence to see that it's true." Here she talked about scientists using evidence to support an idea to see if the idea is true. On the post-interview, she again talked about scientists using evidence as support though her response became more sophisticated. She stated, "To support their main, to support their claim that they have. Cause if they don't have any evidence everybody would just say that they just made it up out of nowhere." Here there is again a focus on support, but a shift to using the language of "claim" as well as the idea that the evidence is not just for the scientist, but to convince other people as well. In both her examples, she focused on the role of evidence is to support the answer to a question, which was the most frequent response provided by the students.

The most prevalent response on the pre-interview about evidence in science class was not "I don't know" in contrast to the responses about explanation and argumentation. Rather, students began the school year by focusing on an *exchange between people*, but by the end of the year focused on the idea that evidence is used to *support an answer to a question*. For example, when talking about science class in his post-interview Sanjay responded, "To support our ideas we have to use evidence from like the text or everything else like we use it to like help us, to always help you when you need to use it, when you need to support it you have to have evidence or it's just going to be a statement." At the end of the school year when talking about evidence in science class, the majority of students talked about how it was used to support an answer. Similar to explanation and argument, students' ideas at the end of the year about science class align more closely to their ideas about scientists.

Data are a key aspect of the definition of evidence, yet the majority of students did not explicitly talk about data in their discussion of evidence when talking about scientists or science class. In terms of scientists, the percentage of students that referenced data remained

Table 8
Students' views about evidence (n = 23)

	Scientists		Science Class		Everyday	
	Pre	Post	Pre	Post	Pre	Post
Link to explanation	9% (2)	4% (1)	4% (1)	4% (1)	0% (0)	9% (2)
Link to argument	9% (2)	13% (3)	4% (1)	0% (0)	9% (2)	9% (2)
Writing	9% (2)	0% (0)	9% (2)	22% (5)	0% (0)	0% (0)
Observation	22% (5)	9% (2)	35% (8)	4% (1)	9% (2)	0% (0)
Exchange between people	48% (11)	30% (7)	43% (10)	26% (6)	70% (16)	70% (16)
Data	43% (10)	43% (10)	26% (6)	48% (11)	57% (13)	65% (15)
Results from an experiment	17% (4)	13% (3)	9% (2)	39% (9)	0% (0)	0% (0)
Support an answer to a question	52% (12)	52% (12)	13% (3)	57% (13)	9% (2)	22% (5)
Other	13% (3)	9% (2)	4% (1)	0% (0)	9% (2)	4% (1)
Don't know	0% (0)	0% (0)	22% (5)	0% (0)	4% (1)	4% (1)

constant from the beginning to the end of the school year. In terms of the science class, the percentage did increase but only half of the students specifically talked about data at the end of the school year. For example, at the end of the school year Kris said the following when talking about what it means for a scientist to use evidence, “Um, that like data, like if, like if you put a fulcrum next, closer to the load like the centimeters, the less, the less effort you have to use.” Although there were some examples of students using the term data or providing concrete examples of data, such as talking about centimeters, Newtons, or other measurements, the majority of students did not use the term or examples in their responses about evidence.

Again, students’ everyday conceptions of evidence remained more constant across the beginning and end of the school year and tended to focus on an *exchange between people*. For example, during both her pre-interview and post-interview Lucia focused on an exchange between people. On the pre-interview she said, “Like if they um, like they don’t believe you or something and then you um, like you can show them what like you did.” Her response focused on an exchange between people where she was trying to convince someone else. Her post-interview included a specific example, but focused on a similar idea. She stated that, “Me and my sister have the same sandals and um, and she, and then we get mixed up sometimes and my sister tells me that it was hers because she has evidence.” In this example, her sister was trying to convince her about who owns the sandals using evidence. Again, she focused on the role of evidence during an exchange between two people. The students’ ideas in the everyday context differed in that they tended to focus on people talking to each other or convincing each other, while when talking about science and science classrooms they focused on using evidence to support an idea, claim, or statement.

Summary of Students’ Ideas. Table 9 presents a summary of the trends in the students’ ideas for explanation, argument, and evidence across the three contexts.

In terms of scientists, students’ ideas shifted when they talked about explanation, but remained more constant when talking about argument and evidence. In terms of science class, this is where the greatest shift occurred in the students’ ideas. Students’ discussion of all three terms shifted over the course of the semester with both explanation and argument moving away from saying “I don’t know.” Although the 5th graders’ ideas align to some degree with how these concepts are discussed in the science education community, there were areas that could be improved such as rarely discussing the role of evidence in argument or the use of data as evidence. In the everyday context, with the exception of talking about argument as an exchange between people, students’ ideas tended to be different than their ideas about scientists or science class. Furthermore, these ideas were more likely to stay constant over the school year.

Students’ Ability to Write Scientific Arguments

Over the course of school year, the 5th grade students’ written scientific arguments improved. First, in terms of the genre of their writing the students’ writing shifted to include a greater focus on argument at the end of the year compared to at the beginning of the year when students writing took on a variety of formats. Figure 2 includes the genre codes for students’ writing from Lessons 1–5, the common writing assignment, Lesson 7 and for the end of the year assessment. Lesson 6 is not included, because students selected a pre-written claim, evidence, and reasoning instead of constructing their own.

In order to be coded as an argument, the students writing needed to include a claim supported with evidence and/or reasoning. With the exception of Lesson 5, students’ writing

Table 9
Summary of students' ideas about explanation, argument, and evidence

	Scientists	Science Class	Everyday
Explanation	Shift Pre: Exchange between people, such as scientists talking to the public Post: How or why a phenomenon occurs, such as how a liquid becomes a solid when it freezes	Shift Pre: Don't know Post: How or why a phenomenon occurs, such as why bees can be found near flowers	More constant Pre: Exchange between people, such as explaining a game to your friends Post: Exchange between people, such as explaining why a person was out in baseball
Argument	More constant Pre: Exchange between people, such as two scientists disagreeing about an explanation Post: Exchange between people, such as scientists disagreeing about what they found out Rarely discussed the role of evidence	Shift Pre: Don't know Post: Exchange between people, such as when someone disagrees with you about the claim Rarely discussed the role of evidence	More constant Pre: Exchange between people, such as getting mad at someone and disagreeing Post: Exchange between people, such as convincing your friends you should go to the park
Evidence	More constant Pre: Support an answer to a question, such as having evidence to find out something and see if it is true Post: Support an answer to a question, such as using evidence to support a claim Rarely discussed the use of data	Shift Pre: Exchange between people, such as having a debate to see who is wrong Post: Support an answer to a question, such as you need evidence to support your answer Rarely discussed the use of data	More constant Pre: Exchange between people, such as if someone does not believe you then you can use evidence Post: Exchange between people, such as using evidence to determine who owns the sandals

included a greater focus on argumentation over the school year. For example, comparing students' first written scientific arguments (Lesson 1) to their last argument in a lesson (Lesson 7) or the end of the year assessment, illustrates that the structure of their writing is considerably different. In Lesson 1, the students writing consisted of 6% personal narrative, 56% informational text, 11% just claim, and 28% argument. The majority of the students wrote scientific information related to the outdoor classroom, but did not provide a claim that explicitly answered the question—How many different habitats are in the outdoor classroom? For example, Damon wrote, “The flowers in the outdoor classroom attracted bees. They sock the polin of the flower. They make honey with that.” He provided a description of the relationship between the flower and bees, but did not specifically talk about the number of habitats in the outdoor classroom. This is in contrast to the last scientific argumentation lesson where 100% of the students' writing was characterized as an argument or on the end of the year assessment where 88% of the students writing was characterized as an argument. In

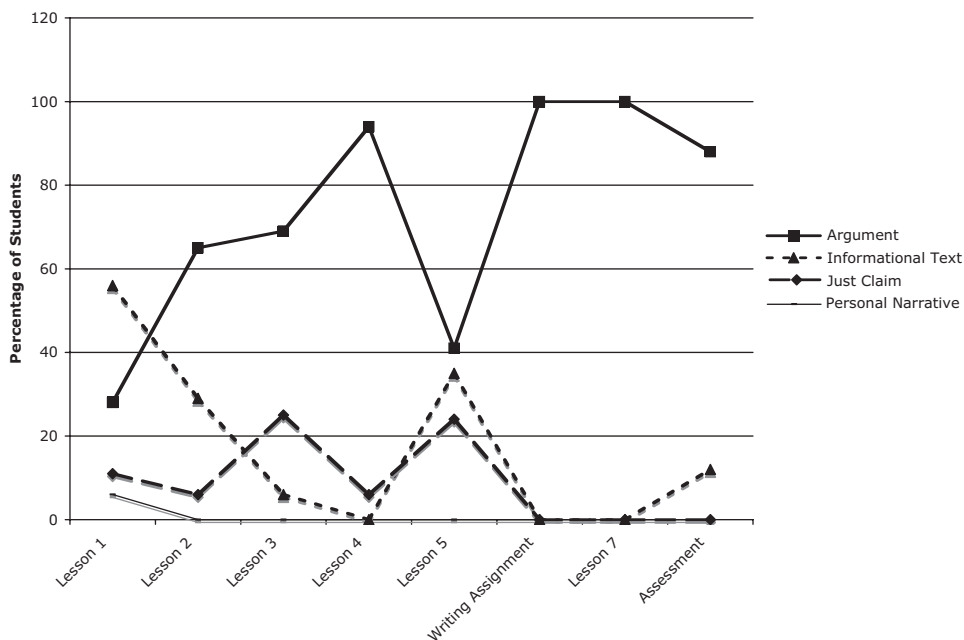


Figure 2. Genre of student writing.

these two examples, students wrote a claim that specifically addressed the question and provided evidence and/or reasoning for that claim. For example, the end of the year assessment asked students to use a food web to construct an argument for the following question—What would happen to the size of the hawk population if all of the seeds were removed from the ecosystem? One student, Liliana, began her response by writing, “The hawk population would decrease because he will barely have food. There will be less hawks because the squirrel and some of the sparrows will die...” In her response, she not only provided a specific claim for the question, but also provided some reasoning and evidence to justify her rationale for why the hawk population would decrease.

Interestingly, in Lesson 5 there was a drop in the percentage of students that wrote arguments (41%) and an increase in the percentage of students who just provided informational text (35%) and did not explicitly address the question or problem asked. This lesson included the most open question of all the lessons in that Mr. Cardone asked his students to review the results from multiple previous investigations during the unit and determine a claim they wanted to make about levers or pulleys. In this lesson, the students had to more specifically define the question for themselves and had to examine multiple datasets, which they found challenging. All of the other writing prompts focused on one investigation or assessment context (Table S2). This suggests that the wording of the writing prompt and the openness of the task can influence the genre of the students’ writing.

In addition to coding for genre, the students’ writing was coded for the quality and scientific accuracy of the arguments in terms of the claim, evidence, and reasoning structure. Figure 3 displays the quality of students’ written arguments for Lessons 1–5, the common writing assignment, Lesson 7, and for the end of the year assessment. As mentioned

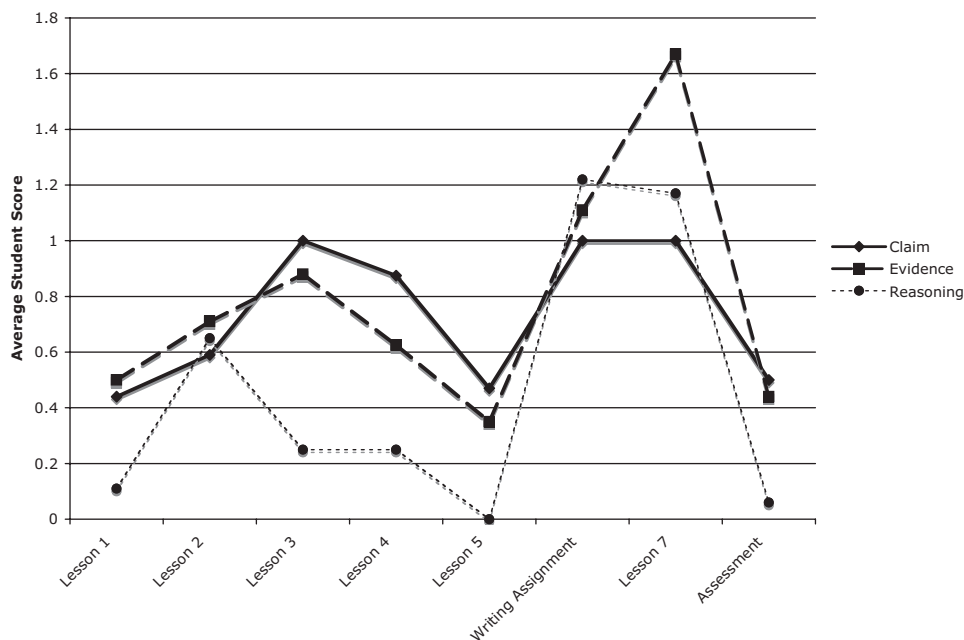


Figure 3. Quality of students' written arguments.

previously, the maximum score for the claim is 1, while for evidence and reasoning the maximum score is 2.

In this coding scheme, the genre of the student writing could be characterized as an argument, but students could still receive a 0 for claim, evidence, and reasoning if their response was not scientifically accurate and appropriate for the question being asked. Although the structure of the students' writing more consistently moved towards argument over the course of the semester, the quality of the students' written arguments fluctuated. For some of the later examples of student writing, such as the common writing assignment about levers and Lesson 7 that focused on pendulums, the quality of the students' written arguments was strong in terms of their ability to construct and justify a scientifically accurate claim. In both of these examples, all of the students received 1 for their claim, because it was scientifically appropriate and accurate and the majority of students included some evidence and reasoning in their writing, with a number of students receiving a 2 for having appropriate and sufficient evidence and reasoning. In contrast, students' written arguments for Lesson 5, in which students specified their own claim from multiple investigations about levers and pulleys, and for the end of the unit assessment about food webs were much weaker. As mentioned earlier, the openness of the question in Lesson 5 was challenging for students and the genre of many of the students' writing did not consist of the structure of an argument. Since the majority of students did not provide a claim to be justified with evidence and/or reasoning, it is not surprising that they also received low scores for quality of the evidence and reasoning. For the end of the year assessment, the majority of students did write an argument, but most of the students' arguments were scientifically inaccurate which is why they received low scores for quality. In contrast to Liliana's argument that included the scientifically accurate claim that the hawk population would decrease, half the students wrote an incorrect claim such as that the hawk population would not change or that all of hawks

would die out. For example, Pedro wrote, “The hawk population will not really change. The evidence I have to prove this is that the squirrel eats the seed and that the sparrow eat the seed but the sparrow can eat the grasshopper so that mean the hawk only loses one animal...” The structure of Pedro’s writing consisted of an argument in that he provided a claim and justified that claim with evidence and reasoning. Yet his understanding of the science behind food webs was incorrect in that the size of the hawk population would change. This example illustrates that understanding the structure of an argument is not sufficient to write a strong scientific argument. Both an understanding of argument and an understanding of the content are necessary for students to write strong scientific arguments.

Discussion

In the context of a learning environment specifically designed to support students in argumentation, both students’ views of scientific explanations, arguments, and evidence changed over the course of the school year and students’ abilities to construct written scientific arguments increased. Explicitly addressing and teaching scientific argumentation over an extended period of time can support students in gaining greater expertise in this complex practice (Osborne et al., 2004). In the case of Mr. Cardone’s class, his 5th grade students developed a stronger understanding of scientific argument yet they still struggled with the accuracy and appropriateness of their scientific arguments when they were asked to write them around challenging content.

Students’ Views of Explanations, Argument, and Evidence

Moje et al. (2001) argue for the importance of classroom interactions that develop students’ awareness of different discourses and connect everyday discourses to the discourses of the science classroom and science community in order to develop congruent third spaces for language, literacy, and science learning in diverse classrooms. The ultimate goal of developing third space in science classrooms is not to teach students that everyday or scientific discourses are right or wrong, but rather to make a space for multiple discourses to help students navigate the hybrid spaces in their lives (Moje et al., 2004). The results from the interviews suggest that, even at the beginning of the year, the 5th grade students were already aware that language could have different meanings in different contexts exemplified by their definitions of explanation, argument, and evidence in the three contexts. Yet the high percentage of students that responded “I don’t know” for the science classroom also suggests that they were unclear of the discourse norms in the science classroom. The students had resources to draw from in both their everyday knowledge and knowledge of the scientists, but were unclear how to use those resources in their science classroom.

Specifically, in the everyday context students’ responses consistently focused on an *exchange between people* when they talked about what it means to create an explanation, create an argument, or use evidence. When talking about explaining, students tended to talk very generally about an exchange between people or they provided a specific example in which they needed to explain how or why a person did something. This differs from an explanation in science, which centers on how or why a natural phenomenon occurs (Berland & Reiser, 2009; Chinn & Brown, 2000; McNeill et al., 2006). In terms of the use of evidence, students again focused on an exchange between people, rather than a more scientific definition such as data that is used to answer a question, solve a problem, or make a decision (Aikenhead, 2005). These everyday practices of explaining and using evidence differed from the scientific practices they were asked to engage in during science class. Consequently, for these two practices, it may have helped students to talk about how their views in their everyday lives

differed from expectations in science class. For argument, students still tended to focus on an exchange between people, but here the examples varied in that they included contexts in which there was a disagreement. This finding differs from Bricker and Bell's (2008) work in which students discussed argument as yelling or fighting in their everyday lives. Yet the finding aligns with Hudicourt-Barnes' (2003) study in which she found in the Haitian culture argument tended to focus on debate in which participants discussed divergent points of view. Although the 5th graders in this study did not specifically discuss debate in reference to argument, they did concentrate on disagreement or the idea that argument occurs when there are multiple different ideas or perspectives. This view of argument in fact aligns with current perspectives of scientific argumentation as a dialogic practice (Jiménez-Aleixandre & Erduran, 2008) and could be a valuable resource to draw from in the science classroom. In terms of classroom discussions, these views of argument could be leveraged to support talk when there are disagreements about science claims that need to be resolved. In terms of writing, scientific argument can again be linked to this idea of an exchange between people or disagreement to support the idea that students should be engaging in persuasive writing in which they should convince the reader of their claim by using appropriate evidence. This alignment of students' views of argumentation is also in contrast to previous concerns that students' may have negative ideas about argumentation (McNeill, 2009). This supports Osborne and Patterson's (2011) case that "argument" may be a more appropriate term than "explanation" for the practice of supporting claims with evidence and reasoning. Students' views of argumentation for their everyday lives and scientists aligned more closely with the practice of argumentation that students should be engaging in during science class.

Over the course of the year, the 5th graders' views of explanation, argument and evidence shifted in terms of the science classroom. Explicitly supporting students through the use of a variety of instructional strategies in developing an understanding of scientific argument allowed the students to develop a stronger understanding of the norms in their science classrooms. Kelly (2005) argues that:

The cultural practices that *count as* science for a group are defined in and through social interaction, including, importantly, uses of language in particular ways for particular purposes. This suggests that in analyzing the educational opportunities for students, educators need to consider the linguistic resources made available and how students are positioned to engage with such resources. (p. 99)

By the end of the school year instead of saying they did not know what these terms meant in science class, the students were explaining them in a similar manner to how they talked about scientists with explanations focusing on how or why phenomena occur, arguments including an exchange between multiple people and evidence supporting an answer to a question or problem. Mr. Cardone's use of the language of claim, evidence, and reasoning as well as the different instructional strategies and activity structures he used throughout the school year acted as linguistic resources to support his students in developing a stronger understanding of what counted as explanation, argument, and evidence in science class.

The instructional practices in Mr. Cardone's classroom align with the recommendations of Nasir et al. (2006) around learning as a cultural process. They argue that in order to support students in recruiting their everyday practices to create meaningful opportunities for academic learning that three different design principles need to be considered: (i) making the structure of the domain visible; (ii) actively engaging students in academic discourses in ways that create meaningful roles and relationships for learners; and (iii) engaging in metalevel analysis that help youth see the relationships between everyday knowledge and

discourse compared to academic knowledge and discourse. All three different design principles were prevalent in the instructional practices of Mr. Cardone's classroom: (i) he used the claim, evidence, and reasoning framework to make the structure of the domain visible; (ii) he frequently engaged his students throughout the school year in both talk and writing around the framework in which students focused on the meaning making of both first hand and second hand experiences; and (iii) in two lessons, he explicitly discussed argumentation in everyday examples and discussed the similarity and differences compared to science. For the everyday examples, he specifically talked about how in an argument you want to convince or persuade someone else. In science class, he wanted the students to do the same thing—convince someone else of their claim during discussions and in their writing. These strategies may have supported students in developing a stronger understanding of explanation, argument, and evidence.

Although the 5th grade students' ideas aligned more closely with scientific perspectives of explanation, argument, and evidence for science and science classrooms at the end of the school year, there were still areas that could have been improved. First of all, in terms of argument at the end of the school year the majority of students did not link argument to the use of evidence either in terms scientists or science class. When considering argument as a structure, one commonality across the many different definitions in the field of science education is the inclusion of data or evidence (Sampson & Clark, 2008). The use of evidence in scientific argumentation is an essential aspect of justifying claims, but not an idea that many students mentioned during their interviews. Instead, the 5th graders focused more on the social or dialogic meaning of argument as the interaction between multiple people (Jiménez-Aleixandre & Erduran, 2008). Second, in students' discussion of evidence they talked about it as supporting an answer to a question in science, but were less likely to talk about evidence as data. Students did not clearly articulate what counted as evidence (e.g., data, measurements, etc.). In looking back at the lessons, although Mr. Cardone explicitly defined evidence as data in Lesson 1, in the majority of the lessons he used more general language in discussing evidence. For example, in the transcript that was included from Lesson 5 he stated, "Step 2 is to provide the evidence for your claim. So here you're going to be using your notes to find that evidence and this is going to be backing up the claim that you state." He never explicitly explained what counted as evidence and he did not emphasize the important role of evidence in argument. Because of these two limitations around evidence and data, during the next school year Mr. Cardone focused more explicitly on discussing the important role of evidence in argument as well as discussing what did and did not count as evidence. The central role of evidence in argumentation also differs from the students' everyday views of argument suggesting that this is a characteristic that should be highlighted in classroom instruction.

One limitation of this study is that the primary data source of students' everyday understandings was one-on-one interviews. The question-and-answer structure, typical of many classroom interactions, may underestimate the everyday resources that students bring to school science (Hudicourt-Barnes, 2003). Other research that follows students into settings outside of the school into their many everyday contexts can better measure the range of resources students bring to science class (Bricker & Bell, 2007). Yet the findings from this study offer some initial promising results. These 5th grade students appeared to successfully differentiate the meaning of the terms evidence, explanation, and argument across the contexts of their everyday lives, their science classrooms, and their views about scientists. Over the course of the year while their views about the terms in science class shifted aligning more closely with accepted academic norms, their everyday views remained more constant. This

suggests that even young students are able to understand that they are navigating discourses in which the meaning of terms can change. Words and phrases have different situated meanings depending upon the context in which they are used (Gee, 2005). Students need to develop flexible understandings of terms and ideas in order to be successful in the many different contexts of their lives.

Students' Written Arguments

Students' ideas about these scientific practices may impact their ability to successfully engage in scientific practices, such as their ability to write scientific arguments. Sandoval (2005) argues that students' practical epistemologies, or their ideas or beliefs about how knowledge claims are produced and evaluated in school science, may impact their ability to engage in the practices of scientific inquiry. The sample size and methods of this study do not support any causal claims about the relationship between students' ideas about argumentation and their abilities to engage in argumentation. However, just as there was a change in students' views about argumentation in science class, there was also a change in students' abilities to write scientific arguments. Future research needs to more explicitly explore the relationship between their views and abilities.

At the beginning of the school year, students struggled with argumentation as a genre of writing. For the majority of students, their initial arguments did not even include a claim that specifically answered the question. Rather, they only provided information related to the topic, such as describing or defining a habitat. Gee (2005) argues that everyday language shares features with other genres, such as poetry, myth, and story telling, which differ greatly from the features of science discourse:

It is typical of everyday language that it tends to obscure the details of causal, or other systematic, relationships among things in favor of rather general and vague relations...Everyday language, in creating patterns and associations, is less careful about differences and underlying systematic relations, though these are crucial to science. (p. 33)

Students' initial writing did not include the relationships or causal patterns in which they justified a claim with evidence and/or reasoning. Rather their initial writing aligned more closely with Gee's (2005) description of more general or vague descriptions about the world around them. When students were asked at the beginning of the year what it means to construct an argument in science class, the majority of students responded that they did not know. This suggests that students were not drawing from their resources about constructing arguments in their everyday lives or when scientists construct arguments, both of which included a focus on an exchange between people and disagreement. If they had drawn from these views, their initial arguments in science class may have looked more persuasive in nature. Rather, the students' writing more closely resembled informational text suggesting they may have been drawing from other genres of writing that they were more familiar with, such as story telling. Providing students with the framework of claim, evidence, and reasoning, may have supported students in understanding this genre of scientific writing and how the structure differed from other forms of writing. By the end of the school year, the structure of the majority of the students' writing did consist of a claim supported with evidence and reasoning.

The elementary students' struggles with constructing an initial claim does differ from work focused on older students where they typically initially provide a claim, but have more difficulty with the justification. Other work I have completed with my colleagues at the

middle school level found that students were able to construct claims before being provided with the claim, evidence, and reasoning framework (McNeill et al., 2006). The framework supported students in justifying those claims, but initially constructing the claims was easier for the older students. This suggests that it may be important to consider a learning progression for scientific argumentation that includes different focuses and support for students across grades k-12 in order to better support students in developing a more sophisticated understanding of argumentation over time (Berland & McNeill, 2010).

Although the genre of the students' writing consistently improved, the students still struggled with the accuracy, appropriateness, and sufficiency of their claims, evidence, and reasoning as they applied the framework to new content areas and contexts. The quality of students' arguments did not show the same consistent pattern of improvement as the genre, but rather fluctuated depending on the specific learning or assessment task. This fluctuation may have occurred for two different reasons: (1) the openness and scaffolding of the questions and (2) students' understanding of the science content. Students' success in constructing scientific arguments is influenced by the wording of the question and the particular dataset (Berland & McNeill, 2010) and the scaffolding and other support provided for the particular question (McNeill & Krajcik, 2009). When questions are more open or include less scaffolding, students can struggle more with the expectations as well as what counts as appropriate evidence and reasoning for a particular claim. Another potential impact on the quality of students' arguments is their understanding of the science content. This result aligns with previous research that suggests the importance of both conceptual knowledge and scientific inquiry practices when engaging in scientific reasoning (Duschl et al., 2007). When experts engage in scientific reasoning, their success is influenced by both domain-general knowledge, such as controlling variables, and domain-specific knowledge, such as an understanding of forces and motion (Schunn & Anderson, 1999). Specifically, in terms of scientific argumentation students' success at writing claims that are appropriately justified is influenced both by their understanding of the structure of an argument and the science concepts (McNeill et al., 2006). If students do not understand the science concepts, they are unable to construct a scientifically accurate claim or appropriately justify that claim. Considering the multiple factors that can influence students' written scientific arguments, it is not surprising that the quality of the students writing did not show a clear linear pathway. The students showed promising growth, yet continued to need support applying the framework to new and more complex contexts and content areas.

Implications

Supporting elementary students in developing proficiency in science in which they are able to use and evaluate scientific evidence, explanations, and arguments is an essential goal of k-8 science. Although these scientific inquiry practices can be challenging for middle school students (McNeill et al., 2006) and high school students (Sandoval, 2003), the results from this study suggest that even elementary students can successfully engage in these practices. In order to be successful in science classrooms, students need to be able to navigate both everyday and scientific discourses (Nasir et al., 2006). Students can have different everyday meanings of these practices compared to the meanings in science (Bricker & Bell, 2008), but this is not a barrier for students' successful participation in those practices. Elementary students are able to understand that "what counts" as an explanation, argument, or evidence in their everyday lives is not the same as "what counts" for scientists and in their science classrooms. All individuals navigate different discourses drawing from multiple resources in those contexts (Moje et al., 2004). Science classrooms need to be designed to utilize those

resources. By providing elementary students with appropriate supports, they can successfully engage in discourse in their science classrooms drawing from their experiences in multiple other contexts.

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